

Software License & Subscription Management System



Design & Implementation of a Relational Database for Centralized License Governance

STUDENT Mihir Kalway	DEPARTMENT SOT	PROFESSOR Dr. Kiran Mayee Adavala	DATE 17th Sept 2025
-------------------------	-------------------	--------------------------------------	------------------------

Licenses expire silently. Seats get over-allocated.



THE PROBLEM

Organizations run many licensed products across departments with no centralized control — licenses expire unnoticed or seats are allocated beyond what was purchased.



SCOPE

In scope — vendors, products, license types, purchases, allocation, subscriptions, renewals, compliance, reporting.

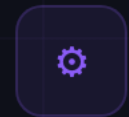
Out of scope — deployment, payment gateways, real-time telemetry.



OBJECTIVES

Normalized 3NF schema with strong integrity constraints. Prevent over-allocation and duplicate keys. Enable expiry, utilization, and compliance reporting. Java/Python front-end for CRUD.

User Roles & Functional Requirements



IT Administrator

Maintains inventory, allocates and reclaims licenses



Department Manager

Requests licenses for team members



Procurement Manager

Records purchases and manages vendor renewals



Compliance Auditor

Runs compliance checks and audit reports



End User

Uses allocated software on their device

— FUNCTIONAL REQUIREMENTS

- ✓ Record vendors, products, purchases with unique license keys
- ✓ Allocate licenses within purchased seat quantity
- ✓ Track subscriptions and process renewals
- ✓ Reclaim licenses when users change roles
- ✓ Run compliance checks against active allocations
- ✓ Generate expiry, utilization, and compliance reports

12 core entities

forming the complete data model

• Vendor

Supplier of software products

• Software Product

Licensed product catalog

• License Type

Perpetual / Sub / Trial / OEM

• Purchase

Purchase order with quantity

• License

Individual license with key

• Subscription

Recurring billing record

• Department

Organizational business unit

• Users

License recipients

• Device

Hardware holding a license

• Allocation

License → user/device link

• Renewal

Extends license validity

• Compliance Check

Periodic audit log